

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Followed PEP 8 standards for Python and Prettier for React.
2	Proper File Structure	Files and folders are logically organized	Yes	Separated into `/backend` (FastAPI), `/frontend` (React), and `/model` (ML artifacts).
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Used descriptive names like `applicant_income` and `approval_probability`.

4	Function/Method Names	Functions are descriptively named	Yes	Functions named clearly, e.g., <code>`predict_approval()`</code> and <code>`preprocess_data()`</code> .
S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
5	Code Comments	Inline and block comments explain logic	Yes	Used docstrings for Python functions and inline comments for complex ML logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Preprocessing, model training, and API routing are in separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Refactored repeated frontend UI components into reusable React components.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Implemented <code>`try-except`</code> blocks in FastAPI and error boundaries in React.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	DataPreprocessor	Python / Pandas	Used in both model training and real-time inference pipelines.	High
2	SHAPExplainer	Python / SHAP	Used in the prediction API to return feature importances for all decisions.	High
3	PredictionCard	JS / React	Reused across the dashboard to display individual applicant results.	High
4	AuthMiddleware	Python / FastAPI	Used across all protected backend API routes for securing endpoints.	High
5	DatabaseConnector	Python / SQLAlchemy	Used for all database CRUD operations and query handling.	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean MVC-like separation between frontend, API, and ML logic.
Readability	5	Strict adherence to PEP 8 conventions and comprehensive docstrings.
Reusability	4	Most components are highly reusable; a few specialized ML tuning scripts are single-use.
Documentation / Comments	5	Included comprehensive README.md, Swagger API docs (via FastAPI), and clear inline comments.
Overall Score	4.75	The codebase is highly maintainable, scalable, well-documented, and ready for deployment.